

COGNIZANT
JAVA FSE ANGULAR

STUDENT COURSE PORTAL

Submitted By

Sk. Waseem Akram

Registration No.: 23BCE7706

SuperSet ID: 8017476

Email: waseem9.money@gmail.com

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Digital Nurture 5.0 - Angular (v20.0)

HANDS-ON 1: Environment Setup, Project Structure & First Component

Task 1: Scaffold and Explore the Angular Project



```
PS C:\Users\wasee> cd student-course-portal
PS C:\Users\wasee\student-course-portal> ng build
Browser bundles
Initial chunk files | Names | Raw size | Estimated transfer size
main-2HZPHX33.js | main | 253.22 kB | 70.72 kB
styles-5INURTS0.css | styles | 0 bytes | 0 bytes
| Initial total | 253.22 kB | 70.72 kB

Server bundles
Initial chunk files | Names | Raw size
server.mjs | server | 816.88 kB
main.server.mjs | main.server | 691.78 kB
polyfills.server.mjs | polyfills.server | 234.18 kB
chunk-IT0R7N0X.mjs | - | 1.38 kB
chunk-KYQQYI4M.mjs | - | 1.01 kB

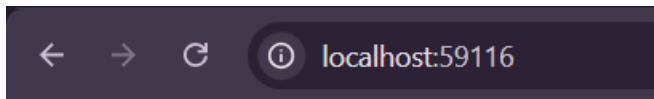
Lazy chunk files | Names | Raw size
chunk-KUERU4XA.mjs | xhr2 | 12.17 kB
chunk-LYHAGGDF.mjs | xhr2 | 12.07 kB

Prerendered 1 static route.
Application bundle generation complete. [4.619 seconds] - 2026-07-18T11:15:15.232Z
Output location: C:\Users\wasee\student-course-portal\dist\student-course-portal
```

```
"budgets": [
  {
    "type": "initial",
    "maximumWarning": "500kb",
    "maximumError": "1mb"
  }
],
```

Task 2: Create and Organise Components

```
PS C:\Users\wasee\student-course-portal> ng generate component components/header
CREATE src/app/components/header/header.spec.ts (554 bytes)
CREATE src/app/components/header/header.ts (197 bytes)
CREATE src/app/components/header/header.css (0 bytes)
CREATE src/app/components/header/header.html (22 bytes)
PS C:\Users\wasee\student-course-portal> ng generate component page/home
CREATE src/app/page/home/home.spec.ts (540 bytes)
CREATE src/app/page/home/home.ts (189 bytes)
CREATE src/app/page/home/home.css (0 bytes)
CREATE src/app/page/home/home.html (20 bytes)
PS C:\Users\wasee\student-course-portal> ng generate component page/course-list
CREATE src/app/page/course-list/course-list.spec.ts (583 bytes)
CREATE src/app/page/course-list/course-list.ts (216 bytes)
CREATE src/app/page/course-list/course-list.css (0 bytes)
CREATE src/app/page/course-list/course-list.html (27 bytes)
PS C:\Users\wasee\student-course-portal> ng generate component page/student-profile
CREATE src/app/page/student-profile/student-profile.spec.ts (611 bytes)
CREATE src/app/page/student-profile/student-profile.ts (232 bytes)
CREATE src/app/page/student-profile/student-profile.css (0 bytes)
CREATE src/app/page/student-profile/student-profile.html (31 bytes)
```



Student Course Portal [Home](#)[Courses](#)[Profile](#)

Welcome

Description

Courses Available: 12 | Enrolled: 3 | GPA: 3.8

HANDS-ON 2: Data Binding, Lifecycle Hooks & Component Communication

Task 1: All Four Binding Types

```
TS home.ts X home.html
src > app > page > home > TS home.ts > Home
1 import { Component } from '@angular/core';
2 import { FormsModule } from '@angular/forms';
3
4 @component({
5   selector: 'app-home',
6   standalone: true,
7   imports: [FormsModule],
8   templateUrl: './home.html',
9   styleUrls: ['./home.css']
10 })
11 export class Home {
12   portalName = 'Student Course Portal';
13
14   isPortalActive = true;
15
16   message = '';
17
18   searchTerm = '';
19
20   // [property] is one-way from component to DOM.
21   // [(ngModel)] is two-way between component and DOM.
22
23   onEnrollClick() {
24     this.message = 'Enrollment opened!';
25   }
26 }
27
```

```
TS home.ts home.html X
src > app > page > home > home.html > p
1 <h1>{{ portalName }}</h1>
2
3 <p>Description</p>
4
5 <div>
6   Courses Available: 12 | Enrolled: 3 | GPA: 3.8
7 </div>
8
9 <br>
10
11 <button
12   [disabled]="!isPortalActive"
13   (click)="onEnrollClick()">
14   Enroll Now
15 </button>
16
17 <p>{{ message }}</p>
18
19 <br>
20
21 <input
22   type="text"
23   [(ngModel)]="searchTerm"
24   placeholder="Search courses">
25
26 <p>Searching for: {{ searchTerm }}</p>
```

← → ↻ 🌐 localhost:63651

Student Course Portal [HomeCoursesProfile](#)

Student Course Portal

Description

Courses Available: 12 | Enrolled: 3 | GPA: 3.8

Enrollment opened!

Searching for: Angular

Task 2: Lifecycle Hooks

```
TS home.ts X
src > app > page > home > TS home.ts > Home
1 import { Component, OnInit, OnDestroy } from '@angular/core';
2 import { FormsModule } from '@angular/forms';
3
4 @Component({
5   selector: 'app-home',
6   standalone: true,
7   imports: [FormsModule],
8   templateUrl: './home.html',
9   styleUrls: ['./home.css']
10 })
11 export class Home implements OnInit, OnDestroy {
12
13   portalName = 'Student Course Portal';
14   isPortalActive = true;
15   message = '';
16   searchTerm = '';
17
18   // [property] is one-way from component to DOM.
19   // [(ngModel)] is two-way between component and DOM.
20
21   ngOnInit() {
22     | console.log('HomeComponent initialised - courses loaded');
23   }
24
25   ngOnDestroy() {
26     | console.log('HomeComponent destroyed');
27   }
28
29   onEnrollClick() {
30     | this.message = 'Enrollment opened!';
31   }
32 }
```

```
TS course-card.ts X
src > app > page > course-card > TS course-card.ts > CourseCard
1 import {
2   Component,
3   Input,
4   Output,
5   EventEmitter,
6   OnChanges,
7   SimpleChanges
8 } from '@angular/core';
9
10 @Component({
11   selector: 'app-course-card',
12   standalone: true,
13   imports: [],
14   templateUrl: './course-card.html',
15   styleUrls: ['./course-card.css']
16 })
17 export class CourseCard implements OnChanges {
18
19   @Input() course: any;
20
21   @Output() enrollRequested = new EventEmitter<number>();
22
23   ngOnChanges(changes: SimpleChanges) {
24     | console.log(
25       | changes['course'].previousValue,
26       | changes['course'].currentValue
27     );
28   }
29 }
```

```
PS C:\Users\wasee\student-course-portal> ng generate component page/course-card
CREATE src/app/page/course-card/course-card.spec.ts (583 bytes)
CREATE src/app/page/course-card/course-card.ts (216 bytes)
CREATE src/app/page/course-card/course-card.css (0 bytes)
CREATE src/app/page/course-card/course-card.html (27 bytes)
```

```
<> course-list.html X
src > app > page > course-list > <> course-list.html >
1 <app-course-card
2   *ngFor="let i of [1,2,3]"
3   [course]="{id:1,name:'Test'}">
4 </app-course-card>
```

```
HomeComponent initialised - courses loaded
HomeComponent destroyed
```

Task 3: @Input and @Output - Parent-Child Communication

```
<> course-card.html X
src > app > page > course-card > <> course-card.html > div.card
1 <div class="card">
2   <h3>{{ course.name }}</h3>
3   <p>ID : {{ course.id }}</p>
4   <p>Code : {{ course.code }}</p>
5   <p>Credits : {{ course.credits }}</p>
6   <button (click)="enrollRequested.emit(course.id)">
7     Enroll
8   </button>
9 </div>
```

```
<> course-list.html X
src > app > page > course-list > <> course-list.html > p
1 <app-course-card
2   *ngFor="let c of courses"
3   [course]="c"
4   (enrollRequested)="onEnroll($event)">
5 </app-course-card>
6
7 <p *ngIf="selectedCourseId">
8   Selected course ID: {{ selectedCourseId }}
9 </p>
```

```
TS course-lists X
src > app > page > course-list > TS course-lists > CourseList
1 import { Component } from '@angular/core';
2 import { CommonModule } from '@angular/common';
3 import { CourseCard } from '../course-card/course-card';
4
5 @Component({
6   selector: 'app-course-list',
7   standalone: true,
8   imports: [CommonModule, CourseCard],
9   templateUrl: './course-list.html',
10  styleUrls: ['./course-list.css']
11 })
12 export class CourseList {
13
14   selectedCourseId = 0;
15
16   courses = [
17     {
18       id: 1,
19       name: 'Data Structures',
20       code: 'CSE201',
21       credits: 4
22     },
23     {
24       id: 2,
25       name: 'Operating Systems',
26       code: 'CSE301',
27       credits: 4
28     },
29     {
30       id: 3,
31       name: 'Computer Networks',
32       code: 'CSE302',
33       credits: 3
34     },
35   ],
```

```
30     {
31       id: 3,
32       name: 'Computer Networks',
33       code: 'CSE302',
34       credits: 3
35     },
36     {
37       id: 4,
38       name: 'DBMS',
39       code: 'CSE303',
40       credits: 4
41     },
42     {
43       id: 5,
44       name: 'Machine Learning',
45       code: 'AIML401',
46       credits: 4
47     }
48   ];
49   onEnroll(courseId: number) {
50     console.log('Enrolling in course: ' + courseId);
51     this.selectedCourseId = courseId;
52   }
53
54 }
```

```
@Input() course!: {
  id: number;
  name: string;
  code: string;
  credits: number;
};

@Output() enrollRequested = new EventEmitter<number>();
```

Student Course Portal [HomeCoursesProfile](#)

Data Structures

ID : 1

Code : CSE201

Credits : 4

Enroll

Operating Systems

ID : 2

Code : CSE301

Credits : 4

Enroll

Computer Networks

ID : 3

Code : CSE302

Credits : 3

Enroll

DBMS

ID : 4

Code : CSE303

Credits : 4

Enroll

Machine Learning

ID : 5

Code : AIML401

Credits : 4

Enroll

Selected course ID: 4

HANDS-ON 3: Directives & Pipes - Built-in and Custom

Task 1: Structural Directives

```
isLoading = true;

ngOnInit() {
  setTimeout(() => {
    this.isLoading = false;
  }, 1500);
}
```

[Student Course Portal](#) [Home](#) [Courses](#) [Profile](#)

Loading courses...

Data Structures

ID: 1

Code: CSE201

Credits: 4 Credits

Passed

[Expand](#) [Enroll](#)

Operating Systems

ID: 2

Code: CSE301

Credits: 4 Credits

Failed

[Expand](#) [Enroll](#)

Computer Networks

ID: 3

Code: CSE302

Credits: 3 Credits

Pending

[Expand](#) [Enroll](#)

Task 2: Attribute Directives ngClass and ngStyle

```
<div
  class="card"
  [ngClass]="cardClasses"
  [style.borderLeftColor]="
    course.gradeStatus === 'passed'
    ? 'green'
    : course.gradeStatus === 'failed'
    ? 'red'
    : 'grey'
">
```

```
get cardClasses(){
  return {
    'card--enrolled': this.isEnrolled,
    'card--full': this.course.credits >= 4,
    'expanded': this.isExpanded
  };
}
```

```
isExpanded = false;

ngOnChanges(changes: SimpleChanges) {
  console.log(
    changes['course']?.previousValue,
    changes['course']?.currentValue
  );
}

toggleExpand(){
  this.isExpanded = !this.isExpanded;
}
```

```
.expanded{
  transform:scale(1.05);
}
.expanded{
  min-height:300px;
  transition:.3s;
}
```

```
<button (click)="toggleExpand()">
  {{ isExpanded ? 'Hide Details' : 'Show Details' }}
</button>
```

Data Structures

ID: 1

Code: CSE201

Credits: 4 Credits

Passed

[Hide Details](#) [Enroll](#)

Operating Systems

ID: 2

Code: CSE301

Credits: 4 Credits

Failed

[Show Details](#) [Enroll](#)

Computer Networks

ID: 3

Code: CSE302

Credits: 3 Credits

Pending

[Show Details](#) [Enroll](#)

Task 3: Custom Directive and Custom Pipe

```
PS C:\Users\wasee\student-course-portal> ng generate directive directives/highlight
CREATE src/app/directives/highlight.spec.ts (207 bytes)
CREATE src/app/directives/highlight.ts (124 bytes)
```

```
PS C:\Users\wasee\student-course-portal> ng generate pipe pipes/credit-label
CREATE src/app/pipes/credit-label-pipe.spec.ts (216 bytes)
CREATE src/app/pipes/credit-label-pipe.ts (240 bytes)
```

```
TS highlight.ts X TS credit-label-pipe.ts TS course-card.ts < >
src > app > directives > TS highlight.ts > ...
1 import {
2   Directive,
3   ElementRef,
4   HostListener,
5   Input
6 } from '@angular/core';
7 @Directive({
8   selector: '[appHighlight]',
9   standalone: true
10 })
11 export class HighlightDirective {
12   @Input() appHighlight = 'yellow';
13   constructor(private el: ElementRef){}
14   @HostListener('mouseenter')
15   mouseEnter(){
16     this.el.nativeElement.style.backgroundColor =
17     this.appHighlight;
18   }
19   @HostListener('mouseleave')
20   mouseLeave(){
21     this.el.nativeElement.style.backgroundColor = '';
22   }
23 }
24
25
26
```

```
src > app > pipes > TS credit-label-pipe.ts > ...
1  import { Pipe, PipeTransform } from '@angular/core';
2  @Pipe({
3    name: 'creditLabel',
4    standalone: true
5  })
6  export class CreditLabelPipe implements PipeTransform {
7    transform(value: number) {
8      return value === 1
9        ? '1 Credit'
10       : value
11        ? value + ' Credits'
12        : 'No Credits';
13    }
14  }
15 }
```

Student Course Portal Home Courses Profile			
Data Structures			
ID:	1		
Code:	CSE201		
Credits:	4 Credits		
Status:	Passed		
		Show Details	Enroll
Operating Systems			
ID:	2		
Code:	CSE301		
Credits:	4 Credits		
Status:	Failed		
		Show Details	Enroll
Computer Networks			
ID:	3		
Code:	CSE302		
Credits:	3 Credits		
Status:	Pending		
		Show Details	Enroll

HANDS-ON 4: Template-Driven Forms & Validation

Task 1: Build the Enrollment Request Form

```
PS C:\Users\wasee\student-course-portal> ng generate component pages/enrollment-form
CREATE src/app/pages/enrollment-form/enrollment-form.spec.ts (611 bytes)
CREATE src/app/pages/enrollment-form/enrollment-form.ts (232 bytes)
CREATE src/app/pages/enrollment-form/enrollment-form.css (0 bytes)
CREATE src/app/pages/enrollment-form/enrollment-form.html (31 bytes)
```

```
TS app.routes.ts 1 X
src > app > TS app.routes.ts > ...
1 import { Routes } from '@angular/router';
2 import { EnrollmentForm } from '../pages/enrollment-form/enrollment-form';
3 import { Home } from '../page/home/home';
4 import { CourseList } from '../page/course-list/course-list';
5 import { StudentProfile } from '../page/student-profile/student-profile';
6
7
8 export const routes: Routes = [
9
10   {
11     path: 'enroll',
12     component: EnrollmentForm
13   }
14 ]
```

```
TS enrollment-form.ts X
src > app > pages > enrollment-form > TS enrollment-form.ts > ...
1 import { Component } from '@angular/core';
2 import { FormsModule } from '@angular/forms';
3
4 @Component({
5   selector: 'app-enrollment-form',
6   standalone: true,
7   imports: [
8     FormsModule
9   ],
10   templateUrl: './enrollment-form.html',
11   styleUrls: ['./enrollment-form.css']
12 })
13 export class EnrollmentForm {
14
15   studentName='';
16   studentEmail='';
17   courseId='';
18   preferredSemester='';
19   agreeToTerms=false;
20
21   onSubmit(form:any){
22
23     console.log(form.value);
24     console.log(form.valid);
25
26   }
27
28 }
29 }
```

```
<> enrollment-form.html X
src > app > pages > enrollment-form > <> enrollment-form.html > ...
1 <h1>Student Enrollment Request</h1>
2 <form #enrollForm="ngForm"(ngSubmit)="onSubmit(enrollForm)">
3   <label>Student Name</label>
4   <input type="text" name="studentName" [(ngModel)]="studentName">
5   <br><br>
6   <label>Student Email</label>
7   <input type="email" name="studentEmail" [(ngModel)]="studentEmail">
8   <br><br>
9   <label>Course ID</label>
10  <input type="number" name="courseId" [(ngModel)]="courseId">
11  <br><br>
12  <label>Preferred Semester</label>
13  <select name="preferredSemester" [(ngModel)]="preferredSemester">
14    <option>Odd</option>
15    <option>Even</option>
16  </select>
17  <br><br>
18  <label>
19    <input type="checkbox" name="agreeToTerms" [(ngModel)]="agreeToTerms">
20    Agree to Terms
21  </label>
22  <br><br>
23  <button type="submit">
24    Submit
25  </button>
26 </form>
```

```
true                                                                    enrollment-form.ts:26
▼ Object ⓘ                                                                enrollment-form.ts:25
  agreeToTerms: true
  courseId: 101
  preferredSemester: "Odd"
  studentEmail: "waseem9.money@gmail.com"
  studentName: "SHAIK"
  ► [[Prototype]]: Object
true                                                                    enrollment-form.ts:26
```

[←](#) [→](#) [↺](#) ⓘ localhost:4200/enroll

Student Course Portal [Home](#) [Courses](#) [Profile](#)

Student Enrollment Request

Student Name

Student Email

Course ID

Preferred Semester

☐ Agree to Terms

Task 2: Validation and Error Messages

```
<input
#nameCtrl="ngModel"
type="text"
name="studentName"
[(ngModel)]="studentName"
required
minlength="3">
<span *ngIf="nameCtrl.touched && nameCtrl.errors?.['required']">
Name is required
</span>
<span *ngIf="nameCtrl.touched && nameCtrl.errors?.['minlength']">
Name must be at least 3 characters
</span>
```

Student Enrollment Request

Student Name : Name must be at least 3 characters

Student Email :

Course ID :

```
<button
type="submit"
[disabled]="enrollForm.invalid">
Submit
</button>

<button
type="button"
(click)="enrollForm.resetForm(); submitted=false">

Reset

</button>

</form>
<div *ngIf="submitted">

Enrollment request submitted successfully!
```

```
export class EnrollmentForm {

  studentName='';
  studentEmail='';
  courseId='';
  preferredSemester='';
  agreeToTerms=false;
  submitted = false;

  onSubmit(form:any){

    console.log(form.value);
    console.log(form.valid);

    if(form.valid){
      | this.submitted = true;
    }

  }

}
```

Student Enrollment Request

Student Name :

Student Email :

Course ID :

Preferred Semester

☒ Agree to Terms

Enrollment request submitted successfully!

HANDS-ON 5: Reactive Forms & Custom Validators

Task 1: Build a Reactive Form with FormBuilder

```
PS C:\Users\wasee\student-course-portal> ng generate component pages/reactive-enrollment-form
CREATE src/app/pages/reactive-enrollment-form/reactive-enrollment-form.spec.ts (668 bytes)
CREATE src/app/pages/reactive-enrollment-form/reactive-enrollment-form.ts (267 bytes)
CREATE src/app/pages/reactive-enrollment-form/reactive-enrollment-form.css (0 bytes)
CREATE src/app/pages/reactive-enrollment-form/reactive-enrollment-form.html (40 bytes)
```

```
export class ReactiveEnrollmentForm implements OnInit {

  enrollForm!: FormGroup;

  constructor(private fb: FormBuilder) {}

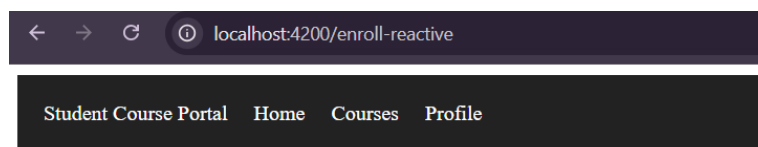
  ngOnInit() {
    this.enrollForm = this.fb.group({
      studentName: [
        '',
        [
          Validators.required,
          Validators.minLength(3)
        ]
      ],
      studentEmail: [
        '',
        [
          Validators.required,
          Validators.email
        ]
      ]
    });
  }
}
```

```
<form
  [formGroup]="enrollForm"
  (ngSubmit)="onSubmit()">
```

```
<input
  type="text"
  FormControlName="studentName">
```

```
<input
  type="email"
  FormControlName="studentEmail"
>
```

```
<input
  type="number"
  FormControlName="courseId"
>
```



Student Enrollment Request (Reactive)

Student Name Name must be at least 3 characters

Student Email

Course ID

Preferred Semester

☒ Agree to Terms


```

<select FormControlName="preferredSemester">
  <option>
    Odd
  </option>
  <option>
    Even
  </option>
</select>

<br><br>

<label>

  <input
    type="checkbox"
    FormControlName="agreeToTerms"
  >

  Agree to Terms

</label>

```

```

onSubmit(){

  console.log("Form Value:");
  console.log(this.enrollForm.value);

  console.log("Raw Value:");
  console.log(this.enrollForm.getRawValue());

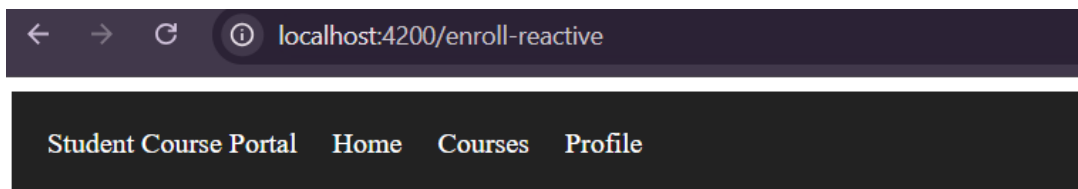
}

```

```

Form Value:
  ▼ Object {
    agreeToTerms: true
    courseId: 101
    preferredSemester: "Odd"
    studentEmail: "waseem9.money@gmail.com"
    studentName: "SHAIK"
    ▶ [[Prototype]]: Object
  }
Raw Value:
  ▼ Object {
    agreeToTerms: true
    courseId: 101
    preferredSemester: "Odd"
    studentEmail: "waseem9.money@gmail.com"
    studentName: "SHAIK"
    ▶ [[Prototype]]: Object
  }

```



Student Enrollment Request (Reactive)

Student Name

Student Email

Course ID

Preferred Semester

☒ Agree to Terms

Task 2: Custom Validators and FormArray for Dynamic Controls

```
<label>
Course ID
</label>

<input
type="text"
formControlName="courseId">

<span
*ngIf="
enrollForm.get('courseId')?.errors?.['noCourseCode']
">

Course code starting with XX is not allowed.

</span>
```

```
TS reactive-enrollment-form.ts ●
src > app > pages > reactive-enrollment-form > TS reactive-enrollment-form.ts > noCourseCode

1 import { Component, OnInit } from '@angular/core';
2 import { CommonModule } from '@angular/common';
3 import {
4   FormBuilder,
5   FormGroup,
6   FormArray,
7   FormControl,
8   AbstractControl,
9   ValidationErrors,
10  Validators,
11  ReactiveFormsModule
12 } from '@angular/forms';
13
14 function noCourseCode(control: AbstractControl): ValidationErrors | null {
15
16   const value = control.value;
17
18   if(value && value.startsWith('XX')){
19     return {
20       noCourseCode:true
21     };
22   }
23
24   return null;
25 }
26
```

```
function simulateEmailCheck(control: AbstractControl): Promise<ValidationErrors | null> {
  return new Promise((resolve)=>{
    setTimeout(()=>{
      if(control.value && control.value.includes('test@')){
        resolve({
          emailTaken:true
        });
      }
      else{
        resolve(null);
      }
    },800);
  });
}
```

Student Enrollment Request (Reactive)

Student Name

Student Email

Course ID Course code starting with XX is not allowed.

Preferred Semester

☐ Agree to Terms

Student Enrollment Request (Reactive)

Student Name

Student Email Email taken

Course ID

Preferred Semester

☒ Agree to Terms

HANDS-ON 6: Services & Dependency Injection

Task 1: Create and Use a Course Service

```
PS C:\Users\wasee\student-course-portal> ng generate service services/course
CREATE src/app/services/course.spec.ts (337 bytes)
CREATE src/app/services/course.ts (86 bytes)
```

```
TS course.ts X
src > app > services > TS course.ts > CourseService
1  import { Injectable } from '@angular/core';
2  import { Course as CourseModel } from 'src/app/models/course.model';
3
4
5  @Injectable({
6    providedIn: 'root',
7  })
8  export class CourseService {}
9
10
11  private courses: CourseModel[] = [
12
13    {
14      id: 1,
15      name: 'Data Structures',
16      code: 'CSE201',
17      credits: 4,
18      gradeStatus: 'passed'
19    },
20
21    {
22      id: 2,
23      name: 'Operating Systems',
24      code: 'CSE301',
25      credits: 4,
26      gradeStatus: 'failed'
27    },
28
29    {
30      id: 3,
31      name: 'Computer Networks',
32      code: 'CSE302',
33      credits: 3,
34      gradeStatus: 'pending'
35    },
36
37    {
38      id: 4,
39      name: 'DBMS',
40      code: 'CSE303',
41      credits: 4,
42      gradeStatus: 'passed'
43    },
44  ],
```

Student Course Portal

Welcome to the Student Course Portal

5

Courses Available

3

Enrolled

3.8

GPA

Enroll Now

Searching for:

```
TS course.model.ts X
src > app > models > TS course.model.ts > ...
1  export interface Course {
2    id: number;
3    name: string;
4    code: string;
5    credits: number;
6    gradeStatus:
7      'passed' | 'failed' | 'pending';
8  }
```

```
PS C:\Users\wasee\student-course-portal> ng generate component components/course-summary-widget
CREATE src/app/components/course-summary-widget/course-summary-widget.spec.ts (647 bytes)
CREATE src/app/components/course-summary-widget/course-summary-widget.ts (255 bytes)
CREATE src/app/components/course-summary-widget/course-summary-widget.css (0 bytes)
CREATE src/app/components/course-summary-widget/course-summary-widget.html (37 bytes)
```

```
import { CourseSummaryWidget }
from 'src/app/components/course-summary-widget/course-summary-widget';
@Component({
  selector: 'app-home',
  standalone: true,
  imports: [FormsModule, CourseSummaryWidget],
```

```
TS course-summary-widget.ts X
src > app > components > course-summary-widget > TS course-summary-widgets >
1 import { Component, OnInit } from '@angular/core';
2 import { CourseService } from '../services/course';
3
4
5 @Component({
6   selector: 'app-course-summary-widget',
7   standalone: true,
8   imports: [],
9   templateUrl: './course-summary-widget.html',
10  styleUrls: ['./course-summary-widget.css']
11 })
12 export class CourseSummaryWidget implements OnInit {
13
14
15   courseCount = 0;
16
17   constructor(
18     | private courseService: CourseService
19   ) {}
20
21
22   ngOnInit(){
23     |
24     | this.courseCount =
25     | this.courseService.getCourses().length;
26   }
27
28 }
29 }
```

Student Course Portal

Welcome to the Student Course Portal

5

Courses Available

3

Enrolled

3.8

GPA

Enroll Now

Search courses

Searching for:

Course Summary

Total Courses: 5

Student Course Portal

Welcome to the Student Course Portal

6

Courses Available

3

Enrolled

3.8

GPA

Enroll Now

Search courses

Searching for:

Course Summary

Total Courses: 6

Task 2: Enrollment Service and Hierarchical DI

```
PS C:\Users\wasee\student-course-portal> ng generate service services/enrollment
CREATE src/app/services/enrollment.spec.ts (357 bytes)
CREATE src/app/services/enrollment.ts (90 bytes)
```

```
enroll(courseId:number):void{
  if(!this.enrolledCourseIds.includes(courseId)){
    this.enrolledCourseIds.push(courseId);
  }
}

unenroll(courseId:number):void{
  this.enrolledCourseIds =
  this.enrolledCourseIds.filter(
    id => id !== courseId
  );
}

isEnrolled(courseId:number):boolean{
  return this.enrolledCourseIds.includes(courseId);
}

getEnrolledCourses():Course[]{
  return this.enrolledCourseIds
    .map(
      id=>this.courseService.getCourseById(id)
    )
    .filter(
      course=>course!==undefined
    ) as Course[];
}
```

```
constructor(
  | private enrollmentService: EnrollmentService
  | ){}

enroll(){
  | if(this.isCourseEnrolled()){
  |   | this.enrollmentService.unenroll(this.course.id);
  | }
  | else{
  |   | this.enrollmentService.enroll(this.course.id);
  | }
  | this.enrollRequested.emit(this.course.id);
  | }
  isCourseEnrolled(){
  | return this.enrollmentService.isEnrolled(
  |   | this.course.id
  | );
  | }
}
```

Enrolled Courses

Operating Systems

Code: CSE301

Credits: 4

Computer Networks

Code: CSE302

Credits: 3

Computer Networks

ID: 3

Code: CSE302

Credits: 3 Credits

Pending

Show Details

Enroll

Computer Networks

ID: 3

Code: CSE302

Credits: 3 Credits

Pending

Show Details

Unenroll

```
PS C:\Users\wasee\student-course-portal> ng generate service services/notification
CREATE src/app/services/notification.spec.ts (367 bytes)
CREATE src/app/services/notification.ts (92 bytes)
PS C:\Users\wasee\student-course-portal> ng generate component components/notification
CREATE src/app/components/notification/notification.spec.ts (596 bytes)
CREATE src/app/components/notification/notification.ts (221 bytes)
CREATE src/app/components/notification/notification.css (0 bytes)
CREATE src/app/components/notification/notification.html (28 bytes)
```

Student Course Portal

Welcome to the Student Course Portal

6

Courses Available

3

Enrolled

3.8

GPA

Enroll Now

Search courses

Searching for:

Course Summary

Total Courses: 6

Notification

Notification service working

HANDS-ON 7: Angular Routing - Guards, Lazy Loading & Route Data

Task 1: Route Configuration, Parameters and Nested Routes

```
PS C:\Users\wasee\student-course-portal> ng generate component pages/courses-layout
CREATE src/app/pages/courses-layout/courses-layout.spec.ts (604 bytes)
CREATE src/app/pages/courses-layout/courses-layout.ts (228 bytes)
CREATE src/app/pages/courses-layout/courses-layout.css (0 bytes)
CREATE src/app/pages/courses-layout/courses-layout.html (30 bytes)

PS C:\Users\wasee\student-course-portal> ng generate component pages/course-detail
CREATE src/app/pages/course-detail/course-detail.spec.ts (597 bytes)
CREATE src/app/pages/course-detail/course-detail.ts (224 bytes)
CREATE src/app/pages/course-detail/course-detail.css (0 bytes)
CREATE src/app/pages/course-detail/course-detail.html (29 bytes)
PS C:\Users\wasee\student-course-portal> ng generate component pages/not-found
CREATE src/app/pages/not-found/not-found.spec.ts (569 bytes)
CREATE src/app/pages/not-found/not-found.ts (208 bytes)
CREATE src/app/pages/not-found/not-found.css (0 bytes)
CREATE src/app/pages/not-found/not-found.html (25 bytes)
```

localhost:4200/fxhdcgfv

Student Course Portal Home Courses Profile

not-found works!

TS app.routes.ts X

src > app > TS app.routes.ts > ...

```
1 import { Routes } from '@angular/router';
2 import { EnrollmentForm } from '../pages/enrollment-form/enrollment-form';
3 import { Home } from '../page/home/home';
4 import { CourseList } from '../page/course-list/course-list';
5 import { StudentProfile } from '../page/student-profile/student-profile';
6 import { ReactiveEnrollmentForm }
7 from '../pages/reactive-enrollment-form/reactive-enrollment-form';
8 import { CourseDetail } from '../pages/course-detail/course-detail';
9 import { NotFound } from '../pages/not-found/not-found';
10 export const routes: Routes = [
11   {
12     path: '',
13     component: Home
14   },
15   {
16     path: 'courses',
17     component: CourseList
18   },
19   {
20     path: 'courses/:id',
21     component: CourseDetail
22   },
23   {
24     path: 'profile',
25     component: StudentProfile
26   },
27   {
28     path: 'enroll',
29     component: EnrollmentForm
30   },
31   {
32     path: 'enroll-reactive',
33     component: ReactiveEnrollmentForm
34   },
35   {
36     path: '**',
37     component: NotFound
38   }
39 ];
```

localhost:4200/courses/5

Student Course Portal Home Courses Profile

Machine Learning

Code: AIML401

Credits: 4

Status: pending

localhost:4200/courses/69

Student Course Portal Home Courses Profile

Course Not Found

localhost:4200/courses?search=ANGULAR

Student Course Portal Home Courses Profile

ANGULAR

Search

Searching for: ANGULAR

Data Structures

ID: 1

Code: CSE201

Credits: 4 Credits

Passed

Show Details Enroll

Task 2: Lazy Loading and Route Guards

```
PS C:\Users\wasee\student-course-portal> ng generate guard guards/auth
? Which type of guard would you like to create?
>Ⓢ CanActivate
  ○ CanActivateChild
  ○ CanDeactivate
  ○ CanMatch
```

```
PS C:\Users\wasee\student-course-portal> ng generate guard guards/auth
✔ Which type of guard would you like to create? CanActivate
CREATE src/app/guards/auth-guard.spec.ts (478 bytes)
CREATE src/app/guards/auth-guard.ts (133 bytes)
PS C:\Users\wasee\student-course-portal> ng generate service services/auth
CREATE src/app/services/auth.spec.ts (327 bytes)
CREATE src/app/services/auth.ts (84 bytes)
```

```
PS C:\Users\wasee\student-course-portal> ng generate guard guards/unsavedchanges
? Which type of guard would you like to create?
  ○ CanActivate
  ○ CanActivateChild
>Ⓢ CanDeactivate
  ○ CanMatch
```

```
PS C:\Users\wasee\student-course-portal> ng generate guard guards/unsavedchanges
✔ Which type of guard would you like to create? CanDeactivate
CREATE src/app/guards/unsavedchanges-guard.spec.ts (531 bytes)
CREATE src/app/guards/unsavedchanges-guard.ts (192 bytes)
```

localhost:4200/enroll-reactive

Student Course Portal Home Courses Profile

Student Enrollment Request (Reactive)

Student Name

Student Email

Course ID

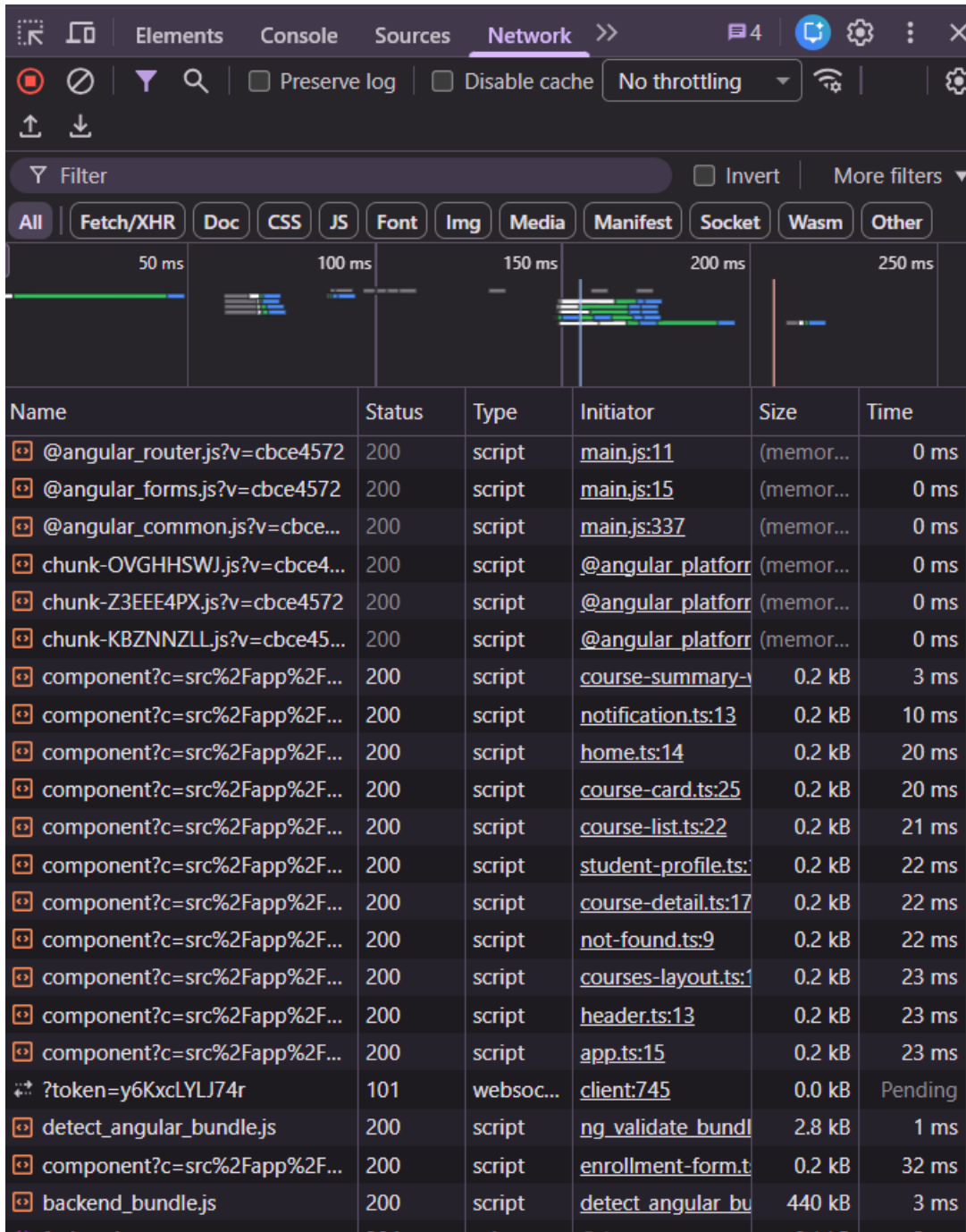
Preferred Semester

☐ Agree to Terms

Additional Courses

localhost:4200 says

You have unsaved changes. Leave?



HANDS-ON 8: HTTP Client - API Integration, Observables & Interceptors

Task 1: Replace Service Data with HttpClient Calls

```
dbjson x
dbjson > ...
1 {
2   "courses": [
3     {
4       "id": 1,
5       "name": "Data Structures",
6       "code": "CSE201",
7       "credits": 4,
8       "gradeStatus": "passed"
9     },
10    {
11      "id": 2,
12      "name": "Operating Systems",
13      "code": "CSE301",
14      "credits": 4,
15      "gradeStatus": "failed"
16    },
17    {
18      "id": 3,
19      "name": "Computer Networks",
20      "code": "CSE302",
21      "credits": 3,
22      "gradeStatus": "pending"
23    },
24    {
25      "id": 4,
26      "name": "DBMS",
27      "code": "CSE303",
28      "credits": 4,
29      "gradeStatus": "passed"
30    },
31    {
32      "id": 5,
33      "name": "Machine Learning",
34      "code": "AIML401",
35      "credits": 4,
36      "gradeStatus": "pending"
37    },
38    {
39      "id": 6,
40      "name": "Artificial Intelligence",
41      "code": "AIML402",
42      "credits": 4,
43      "gradeStatus": "passed"
44    }
45  ]
46 }
```

```
localhost:3000/courses
Pretty print
[
  {
    "id": "1",
    "name": "Data Structures",
    "code": "CSE201",
    "credits": 4,
    "gradeStatus": "passed"
  },
  {
    "id": "2",
    "name": "Operating Systems",
    "code": "CSE301",
    "credits": 4,
    "gradeStatus": "failed"
  },
  {
    "id": "3",
    "name": "Computer Networks",
    "code": "CSE302",
    "credits": 3,
    "gradeStatus": "pending"
  },
  {
    "id": "4",
    "name": "DBMS",
    "code": "CSE303",
    "credits": 4,
    "gradeStatus": "passed"
  },
  {
    "id": "5",
    "name": "Machine Learning",
    "code": "AIML401",
    "credits": 4,
    "gradeStatus": "pending"
  },
  {
    "id": "6",
    "name": "Artificial Intelligence",
    "code": "AIML402",
    "credits": 4,
    "gradeStatus": "passed"
  }
]
```

```
import { HttpClient } from '@angular/common/http';

@Injectable({
  providedIn: 'root',
})
export class CourseService {
  constructor(
    private http: HttpClient
  ) {}
}
```

```
export class CourseService {
  private apiUrl = 'http://localhost:3000/courses';
  constructor(
    | private http: HttpClient
  ){}

  getCourses(): Observable<CourseModel[]> {
    | return this.http.get<CourseModel[]>(this.apiUrl);
  }

  getCourseById(id:number): Observable<CourseModel>{
    | return this.http.get<CourseModel>(
    | `_${this.apiUrl}/${id}`
    | );
  }
}
```

```
ngOnInit(){
  this.courseService.getCourses()
    .subscribe({
      next: (courses) => {
        | this.courses = courses;
      },
      error: (err) => {
        | console.log(err);
      },
      complete: () => {
        | this.isLoading = false;
      }
    });
}
```

```
PS C:\Users\wasee\student-course-portal> json-server db.json --port 3000
JSON Server started on PORT :3000
Press CTRL-C to stop
Watching db.json...

( - ^ ▢ ^ - )

Index:
http://localhost:3000/

Static files:
Serving ./public directory if it exists

Endpoints:
http://localhost:3000/courses
http://localhost:3000/students
http://localhost:3000/enrollments
```

SHAIK

ID: hT0HzPFz9XM

Code: 1

Credits: 3 Credits

Pending

Show Details Enroll

Manage Courses

Data Structures (CSE201)

Update Delete

Operating Systems (CSE301)

Update Delete

Computer Networks (CSE302)

Update Delete

DBMS (CSE303)

Update Delete

Machine Learning (AIML401)

Update Delete

Artificial Intelligence (AIML402)

Update Delete

SHAIK (1)

Update Delete

Task 2: RxJS Operators and Error Handling

Student Enrollment Request

Student Name :

Student Email :

Course ID :

Preferred Semester

☒ Agree to Terms

Enrollment request submitted successfully!

```
Courses loaded: 6
course.ts:26
undefined
course-card.ts:63
  ▶ {id: '1', name: 'Data Structures', code: 'CSE201', credits: 4, gradeStatus: 'passed'}
undefined
course-card.ts:63
  ▶ {id: '2', name: 'Operating Systems', code: 'CSE301', credits: 4, gradeStatus: 'failed'}
undefined
course-card.ts:63
  ▶ {id: '3', name: 'Computer Networks', code: 'CSE302', credits: 3, gradeStatus: 'pending'}
undefined
course-card.ts:63
  ▶ {id: '4', name: 'DBMS', code: 'CSE303', credits: 4, gradeStatus: 'passed'}
undefined
course-card.ts:63
  ▶ {id: '5', name: 'Machine Learning', code: 'AIML401', credits: 4, gradeStatus: 'passed'}
undefined
course-card.ts:63
  ▶ {id: '6', name: 'Artificial Intelligence', code: 'AIML402', credits: 4, gradeStatus: 'passed'}
```

```
getCourses(): Observable<CourseModel[]> {
  return this.http
    .get<CourseModel[]>(this.apiUrl)
    .pipe(
      tap(courses => {
        console.log(
          'Courses loaded:',
          courses.length
        );
      }),
      map(courses =>
        courses.filter(
          course => course.credits > 0
        )
      ),
      catchError(err => {
        console.error(err);
        return throwError(
          () => new Error(
            'Failed to load courses. Please try again.'
          )
        );
      })
    );
}
```

```
getCourses(): Observable<CourseModel[]> {
  return this.http
    .get<CourseModel[]>(this.apiUrl)
    .pipe(
      tap(courses => {
        console.log(
          'Courses loaded:',
          courses.length
        );
      }),
      map(courses =>
        courses.filter(
          course => course.credits > 0
        )
      ),
      retry(2),
      catchError(err => {
        console.error(err);
        return throwError(
          () => new Error(
            'Failed to load courses. Please try again.'
          )
        );
      })
    );
}
```

```
PS C:\Users\waseem\student-course-portal> ng generate service services/student
CREATE src/app/services/student.spec.ts (342 bytes)
CREATE src/app/services/student.ts (87 bytes)
```

SHAIK

ID: hT0HzPFz9XM

Code: 1

Credits: 3 Credits

Pending

Show Details Enroll

Manage Courses

Data Structures (CSE201)

View Students Update Delete

Operating Systems (CSE301)

View Students Update Delete

Computer Networks (CSE302)

View Students Update Delete

DBMS (CSE303)

View Students Update Delete

Machine Learning (AIML401)

View Students Update Delete

Artificial Intelligence (AIML402)

View Students Update Delete

Selected course ID: 1

Students Enrolled

Rahul

Amit

Manage Courses

Data Structures (CSE201)

Update Delete

Operating Systems (CSE301)

Update Delete

Computer Networks (CSE302)

Update Delete

DBMS (CSE303)

Update Delete

Machine Learning (AIML401)

Update Delete

Artificial Intelligence (AIML402)

Update Delete

SHAIK (1)

Update Delete

Task 3: HTTP Interceptors

```
PS C:\Users\wasee\student-course-portal> ng generate interceptor interceptors/auth
CREATE src/app/interceptors/auth-interceptor.spec.ts (493 bytes)
CREATE src/app/interceptors/auth-interceptor.ts (154 bytes)
```

```
PS C:\Users\wasee\student-course-portal> ng generate interceptor interceptors/error-handler
CREATE src/app/interceptors/error-handler-interceptor.spec.ts (526 bytes)
CREATE src/app/interceptors/error-handler-interceptor.ts (162 bytes)
PS C:\Users\wasee\student-course-portal> ng generate service services/loading
CREATE src/app/services/loading.spec.ts (342 bytes)
CREATE src/app/services/loading.ts (87 bytes)
PS C:\Users\wasee\student-course-portal> ng generate interceptor interceptors/loading
CREATE src/app/interceptors/loading-interceptor.spec.ts (505 bytes)
CREATE src/app/interceptors/loading-interceptor.ts (157 bytes)
```

```
TS auth-interceptor.ts X TS app.config.ts
src > app > interceptors > TS auth-interceptor.ts > ...
1 import { HttpInterceptorFn } from '@angular/common/http';
2
3
4 export const authInterceptor: HttpInterceptorFn = (req, next) => {
5
6
7     const modifiedRequest = req.clone({
8
9         setHeaders: {
10
11             Authorization: 'Bearer mocktoken-12345'
12         }
13     });
14
15     return next(modifiedRequest);
16
17
18
19
20
21 };
```

```
TS auth-interceptor.ts TS app.config.ts X
src > app > TS app.config.ts > ...
1 import { ApplicationConfig, provideBrowserGlobalErrorListeners } from '@angular/
2 import { provideRouter } from '@angular/router';
3 import { provideHttpClient, withInterceptors } from '@angular/common/http';
4
5 import { routes } from './app.routes';
6 import { provideClientHydration } from '@angular/platform-browser';
7 import { authInterceptor } from './interceptors/auth-interceptor';
8
9
10 export const appConfig: ApplicationConfig = {
11
12     providers:[
13
14         provideBrowserGlobalErrorListeners(),
15
16         provideRouter(routes),
17
18         provideHttpClient(
19             withInterceptors([
20                 authInterceptor
21             ])
22         ),
23
24         provideClientHydration()
25
26     ]
27
28 };
```

```
TS error-handler-interceptor.ts X
src > app > interceptors > TS error-handler-interceptor.ts > ...
1 import { HttpInterceptorFn } from '@angular/common/http';
2 import { catchError, throwError } from 'rxjs';
3 export const errorHandlerInterceptor: HttpInterceptorFn =
4 (req, next) => {
5     return next(req).pipe(
6         catchError(error=>{
7             if(error.status === 401){
8                 console.log(
9                     'Unauthorized'
10                );
11            }
12            if(error.status === 500){
13                console.log(
14                    'Server error'
15                );
16            }
17            return throwError(
18                ()=>error
19            );
20        })
21    );
22 };
```

```
TS loading.ts X
src > app > services > TS loading.ts > ...
1 import { Injectable } from '@angular/core';
2 import { BehaviorSubject } from 'rxjs';
3 @Injectable({
4     providedIn: 'root'
5 })
6 export class LoadingService {
7     private loading =
8     new BehaviorSubject<boolean>(false);
9     isLoading$ =
10     this.loading.asObservable();
11     show(){
12         this.loading.next(true);
13     }
14     hide(){
15         this.loading.next(false);
16     }
17 }
```

Courses

Loading courses...

Failed to load courses. Please try again.

Search

Searching for:

No courses available.

Students Enrolled

Select a course to view students.

3G ▾

Disabled

No throttling

Presets

Fast 4G

Slow 4G

3G

Offline

Custom

Add...

HANDS-ON 9: State Management - NgRx, Actions, Reducers, Effects & Selectors

Task 1: Set Up NgRx Store and Define Course State

```
PS C:\Users\wasee\student-course-portal> npm install @ngrx/store@21.1.1 @ngrx/effects@21.1.1 @ngrx/entity@21.1.1 @ngrx/store-devtools@21.1.1 --legacy-peer-deps

added 4 packages, removed 1 package, and audited 479 packages in 5s

110 packages are looking for funding
  run `npm fund` for details

5 vulnerabilities (1 low, 3 moderate, 1 high)

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
npm warn install-scripts 5 packages had install scripts blocked because they are not covered by allowScripts:
npm warn install-scripts @parcel/watcher@2.5.6 (install: node-gyp rebuild)
npm warn install-scripts esbuild@0.28.1 (postinstall: node install.js)
npm warn install-scripts lmdb@3.5.4 (install: node-gyp rebuild)
npm warn install-scripts msgpackr-extract@3.0.4 (install: node-gyp rebuild)
npm warn install-scripts esbuild@0.27.7 (postinstall: node install.js)
npm warn install-scripts Run `npm install-scripts ls` to review, or `npm install-scripts approve <pkg>` to allow.
PS C:\Users\wasee\student-course-portal> npm list @ngrx/store
student-course-portal@0.0.0 C:\Users\wasee\student-course-portal
+-- @ngrx/effects@21.1.1
|   -- @ngrx/store@21.1.1 deduped
+-- @ngrx/entity@21.1.1
|   -- @ngrx/store@21.1.1 deduped
+-- @ngrx/store-devtools@21.1.1
|   -- @ngrx/store@21.1.1 deduped
+-- @ngrx/store@21.1.1
```

```
PS C:\Users\wasee\student-course-portal> mkdir src\app\store
```

Directory: C:\Users\wasee\student-course-portal\src\app

Mode	LastWriteTime	Length	Name
d-----	24-07-2026 05:29		store

```
PS C:\Users\wasee\student-course-portal> mkdir src\app\store\course
```

Directory: C:\Users\wasee\student-course-portal\src\app\store

Mode	LastWriteTime	Length	Name
d-----	24-07-2026 05:29		course

```
PS C:\Users\wasee\student-course-portal> mkdir src\app\store\enrollment
```

Directory: C:\Users\wasee\student-course-portal\src\app\store

Mode	LastWriteTime	Length	Name
d-----	24-07-2026 05:29		enrollment

```

TS course.effects.ts X
src > app > store > course > TS course.effects.ts > ...
1 import { Injectable } from '@angular/core';
2 import { Actions, createEffect, ofType } from '@ngrx/effects';
3 import { CourseService } from '../../services/course';
4 import {
5   loadCourses,
6   loadCoursesSuccess,
7   loadCoursesFailure
8 } from './course.actions';
9 import {
10   catchError,
11   map,
12   switchMap,
13   of
14 } from 'rxjs';
15 @Injectable()
16 export class CourseEffects {
17   constructor(
18     private actions$: Actions,
19     private courseService: CourseService
20   ) {}
21   loadCourses$ = createEffect(() =>
22     this.actions$.pipe(
23       ofType(loadCourses),
24       switchMap(() =>
25         this.courseService
26           .getCourses()
27           .pipe(
28             map(courses =>
29               loadCoursesSuccess({
30                 courses
31               })
32             ),
33             catchError(error =>
34               of(
35                 loadCoursesFailure({
36                   error: error.message
37                 })
38               )
39             )
40           )
41       )
42     );
43   );
44 }

```

```

TS course.reducer.ts X
src > app > store > course > TS course.reducer.ts > ...
1 import { createReducer, on } from '@ngrx/store';
2 import { Course } from '../../models/course.model';
3 import {
4   loadCourses,
5   loadCoursesSuccess,
6   loadCoursesFailure
7 } from './course.actions';
8 export interface CourseState {
9   courses: Course[];
10  loading: boolean;
11  error: string | null;
12 }
13 export const initialState: CourseState = {
14   courses: [],
15   loading: false,
16   error: null
17 };
18 export const courseReducer = createReducer(
19   initialState,
20   on(loadCourses, (state) => ({
21     ...state,
22     loading: true,
23     error: null
24   })),
25   on(loadCoursesSuccess, (state, { courses }) => ({
26     ...state,
27     courses,
28     loading: false
29   })),
30   on(loadCoursesFailure, (state, { error }) => ({
31     ...state,
32     loading: false,
33     error
34   })))
35 );

```

```

TS course.selectors.ts X
src > app > store > course > TS course.selectors.ts > ...
1 import { createFeatureSelector, createSelector } from '@ngrx/store';
2 import { CourseState } from './course.reducer';
3 // Get course feature state
4 export const selectCourseState =
5   createFeatureSelector<CourseState>('course');
6 // Get all courses
7 export const selectAllCourses =
8   createSelector(
9     selectCourseState,
10    (state) => state.courses
11  );
12 // Get loading status
13 export const selectCoursesLoading =
14   createSelector(
15     selectCourseState,
16    (state) => state.loading
17  );
18 // Get error message
19 export const selectCoursesError =
20   createSelector(
21     selectCourseState,
22    (state) => state.error
23  );

```

```

TS course.actions.ts X
src > app > store > course > TS course.actions.ts > ...
1 import { createAction, props } from '@ngrx/store';
2 import { Course } from '../../models/course.model';
3
4
5 export const loadCourses = createAction(
6   '[Course] Load Courses'
7 );
8
9
10 export const loadCoursesSuccess = createAction(
11   '[Course] Load Courses Success',
12   props<{ courses: Course[] }>()
13 );
14
15
16 export const loadCoursesFailure = createAction(
17   '[Course] Load Courses Failure',
18   props<{ error: string }>()
19 );

```

Task 2: NgRx Effects for HTTP and Enrollment State

The screenshot shows a web application at `localhost:4200/courses` with a navigation bar containing "Student Course Portal", "Home", "Courses", and "Profile". The main content area is titled "Courses" and has a search bar. Below the search bar, there are two course cards:

- Data Structures**
ID: 1
Code: CSE201
Credits: 4 Credits
Status: Passed
Buttons: Show Details, Enroll
- Operating Systems**
ID: 2
Code: CSE301
Credits: 4 Credits
Status: Failed
Buttons: Show Details, Enroll

Overlaid on the right is the Redux DevTools interface. The "Actions" tab shows a list of actions: "[Course] Load Courses", "Success", "[Course] Load Courses", "Success", "[Course] Load Courses", "Success", and "[Course] Load Courses", "Success". The "Diff" tab shows the state changes, with a tree view showing `course (pin)` and `loading (pin): true => false`. The "Inspector" tab shows the current state of the application.

```
TS enrollment.actions.ts X
src > app > store > enrollment > TS enrollment.actions.ts > ...
1 import { createAction, props } from '@ngrx/store';
2
3
4 export const enrollInCourse = createAction(
5   | '[Enrollment] Enroll In Course',
6   props<{ courseId: number }>()
7 );
8
9
10 export const unenrollFromCourse = createAction(
11   | '[Enrollment] Unenroll From Course',
12   props<{ courseId: number }>()
13 );
14
15
16 export const setEnrolledCourses = createAction(
17   | '[Enrollment] Set Enrolled Courses',
18   props<{ courseIds: number[] }>()
19 );
```

```
TS enrollment.selectors.ts X
src > app > store > enrollment > TS enrollment.selectors.ts > ...
1 import { createFeatureSelector, createSelector } from '@ngrx/store';
2 import { EnrollmentState } from './enrollment.reducer';
3 import { selectAllCourses } from './course/course.selectors';
4 // Get enrollment state
5 export const selectEnrollmentState =
6   | createFeatureSelector<EnrollmentState>('enrollment');
7 // Get enrolled course ids
8 export const selectEnrolledIds =
9   | createSelector(
10     selectEnrollmentState,
11     (state) => state.enrolledCourseIds
12   );
13 // Cross-slice selector
14 // Combines enrollment state + course state
15 export const selectEnrolledCourses =
16   | createSelector(
17     selectAllCourses,
18     selectEnrolledIds,
19     (courses, enrolledIds) =>
20       | courses.filter(
21         | course => enrolledIds.includes(course.id)
22       )
23   );
```

The screenshot shows the Redux DevTools interface with the "Actions" tab selected. The list of actions includes:

- `@ngrx/effects/init`
- `[Course] Load Courses`
- `[Course] Load Courses`
- `Success`
- `[Enrollment] Enroll In Course`
- `[Enrollment] Unenroll From Course`

The "Diff" tab shows the state changes, with a tree view showing `enrollment (pin)` and `enrolledCourseIds (pin): { 0: "1" }`. The "Inspector" tab shows the current state of the application.

```
TS enrollment.reducer.ts X
src > app > store > enrollment > TS enrollment.reducer.ts > ...
1 import { createReducer, on } from '@ngrx/store';
2 import {
3   enrollInCourse,
4   unenrollFromCourse,
5   setEnrolledCourses
6 } from './enrollment.actions';
7 export interface EnrollmentState {
8   enrolledCourseIds: number[];
9 }
10 export const initialState: EnrollmentState = {
11   enrolledCourseIds: []
12 };
13 export const enrollmentReducer = createReducer(
14   initialState,
15   on(
16     enrollInCourse,
17     (state, { courseId }) => ({
18       ...state,
19       enrolledCourseIds: [
20         ...state.enrolledCourseIds,
21         courseId
22       ]
23     })
24   ),
25   on(
26     unenrollFromCourse,
27     (state, { courseId }) => ({
28       ...state,
29       enrolledCourseIds:
30         state.enrolledCourseIds.filter(
31           id => id !== courseId
32         )
33     })
34   ),
35   on(
36     setEnrolledCourses,
37     (state, { courseIds }) => ({
38       ...state,
39       enrolledCourseIds: courseIds
40     })
41   )
42 );
```

HANDS-ON 10: Unit Testing Angular Applications - Jasmine, Karma & TestBed

Task 1: Testing a Component – CourseCardComponent

```
TS course-card.spec.ts X
src > app > page > course-card > TS course-card.spec.ts > describe('CourseCard', () => {
10 describe('CourseCard', () => {
25   beforeEach(async () => {
26     await TestBed.configureTestingModule({
27       imports: [
28         CourseCard
29       ],
30       providers: [
31         provideMockStore({
32           initialState: {
33             enrollment: {
34               enrolledCourseIds: []
35             }
36           }
37         })
38       ]
39     }).compileComponents();
40   });
41   vi.spyOn(component.enrollRequested, 'emit');
42   vi.spyOn(console, 'log');
43   it('should create', () => {
44     expect(component).toBeTruthy();
45   });
46 });
47
48
49
50
51
```

```
vi.spyOn(
  component.enrollRequested,
  'emit'
);
```

```
vi.spyOn(
  console,
  'log'
);
```

```
expect(
  component.enrollRequested.emit
).toHaveBeenCalledTimes(1);
```

```
TS course-card.spec.ts
src > app > page > course-card > TS course-card.spec.ts > describe('CourseCard', () => {
10 describe('CourseCard', () => {
25   beforeEach(async () => {
26     await TestBed.configureTestingModule({
27       imports: [
28         CourseCard
29       ],
30       providers: [
31         provideMockStore({
32           initialState: {
33             enrollment: {
34               enrolledCourseIds: []
35             }
36           }
37         })
38       ]
39     }).compileComponents();
40   });
41   fixture = TestBed.createComponent(CourseCard);
42   component = fixture.componentInstance;
43   it('should create', () => {
44     expect(component).toBeTruthy();
45   });
46 });
47
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
```

```
component.ngOnChanges({
  course: {
    previousValue: null,
    currentValue: mockCourse,
    firstChange: true,
    isFirstChange: () => true
  }
});
```

Task 2: Testing a Service and an NgRx-Connected Component

```
service.getCourses()

.subscribe(courses => {

  expect(courses.length)
  .toBe(2);

});

const request =
  httpMock.expectOne(
    | 'http://localhost:3000/courses'
  );
```

```
it('should handle server error', () => {

  service.getCourses()

  .subscribe({

    next: () => {

      throw new Error('Expected error');

    },

    error: (error) => {

      expect(error.message)
      .toBe(
        | 'Failed to load courses. Please try again.'
      );

    }

  });

});
```

```
request.flush(

  'Server error',

  {

    status: 500,
    statusText: 'Server Error'

  }

);
```

```
service =
  TestBed.inject(CourseService);

httpMock =
  TestBed.inject(HttpTestingController);
```

```
providers:[

  provideMockStore({

    initialState:{

      course:{

        courses:[

          {

            id:1,
            name:'Data Structures',
            code:'CS101',
            credits:4,
            gradeStatus:'passed'

          }

        ],
        loading:false,
        error:null

      }

    }

  })

]
```

```
it('should create',()=>{

  expect(component)
  .toBeTruthy();

});

it('should show loading indicator',()=>{

  store.setState({

    course:{

      courses:[],

      loading:true,

      error:null

    }

  });

});
```

```

PS C:\Users\wasee\student-course-portal> ng test
Application bundle generation complete. [1.263 seconds] - 2026-07-24T02:42:39.391Z

Watch mode enabled. Watching for file changes...

DEV v4.1.10 C:/Users/wasee/student-course-portal

✓ student-course-portal src/app/page/student-profile/student-profile.spec.ts (1 test) 56ms
stdout | src/app/components/course-summary-widget/course-summary-widget.spec.ts > CourseSummaryWidget > should create
Courses loaded: 6

✓ student-course-portal src/app/components/course-summary-widget/course-summary-widget.spec.ts (1 test) 70ms
✓ student-course-portal src/app/pages/not-found/not-found.spec.ts (1 test) 10ms
stdout | src/app/services/course.spec.ts > CourseService > should get courses
Courses loaded: 2

stderr | src/app/services/course.spec.ts > CourseService > should handle server error
HttpResponse {
  headers: HttpHeaders {
    headers: Map(0) {},
    normalizedNames: Map(0) {},
    lazyInit: undefined,
    lazyUpdate: null
  },
  status: 500,
  statusText: 'Server Error',
  url: 'http://localhost:3000/courses',
  ok: false,
  type: undefined,
  redirected: undefined,
  responseType: undefined,
  name: 'HttpResponse',
  message: 'Http failure response for http://localhost:3000/courses: 500 Server Error',
  error: 'Server error'
}

```

```

✓ student-course-portal src/app/services/course.spec.ts (3 tests) 9ms
✓ student-course-portal src/app/components/notification/notification.spec.ts (1 test) 8ms
✓ student-course-portal src/app/components/header/header.spec.ts (1 test) 65ms
✓ student-course-portal src/app/app.spec.ts (1 test) 84ms
✓ student-course-portal src/app/pages/reactive-enrollment-form/reactive-enrollment-form.spec.ts (1 test) 83ms
✓ student-course-portal src/app/interceptors/auth-interceptor.spec.ts (1 test) 1ms
✓ student-course-portal src/app/pipes/credit-label-pipe.spec.ts (1 test) 1ms
✓ student-course-portal src/app/pages/enrollment-form/enrollment-form.spec.ts (1 test) 99ms
stdout | src/app/page/home/home.spec.ts > Home > should create
HomeComponent initialised - courses loaded

✓ student-course-portal src/app/pages/courses-layout/courses-layout.spec.ts (1 test) 26ms
✓ student-course-portal src/app/interceptors/loading-interceptor.spec.ts (1 test) 1ms
✓ student-course-portal src/app/interceptors/error-handler-interceptor.spec.ts (1 test) 1ms
stdout | src/app/page/home/home.spec.ts > Home > should create
Courses loaded: 6

stdout | src/app/page/home/home.spec.ts > Home > should create
Courses loaded: 6

stdout | src/app/page/home/home.spec.ts > Home > should create
HomeComponent destroyed

✓ student-course-portal src/app/page/home/home.spec.ts (1 test) 127ms
✓ student-course-portal src/app/pages/course-detail/course-detail.spec.ts (1 test) 6ms
stdout | src/app/page/course-card/course-card.spec.ts > CourseCard > should test ngOnChanges
null {
  id: 1,
  name: 'Data Structures',
  code: 'CS101',
  credits: 4,
  gradeStatus: 'passed'
}

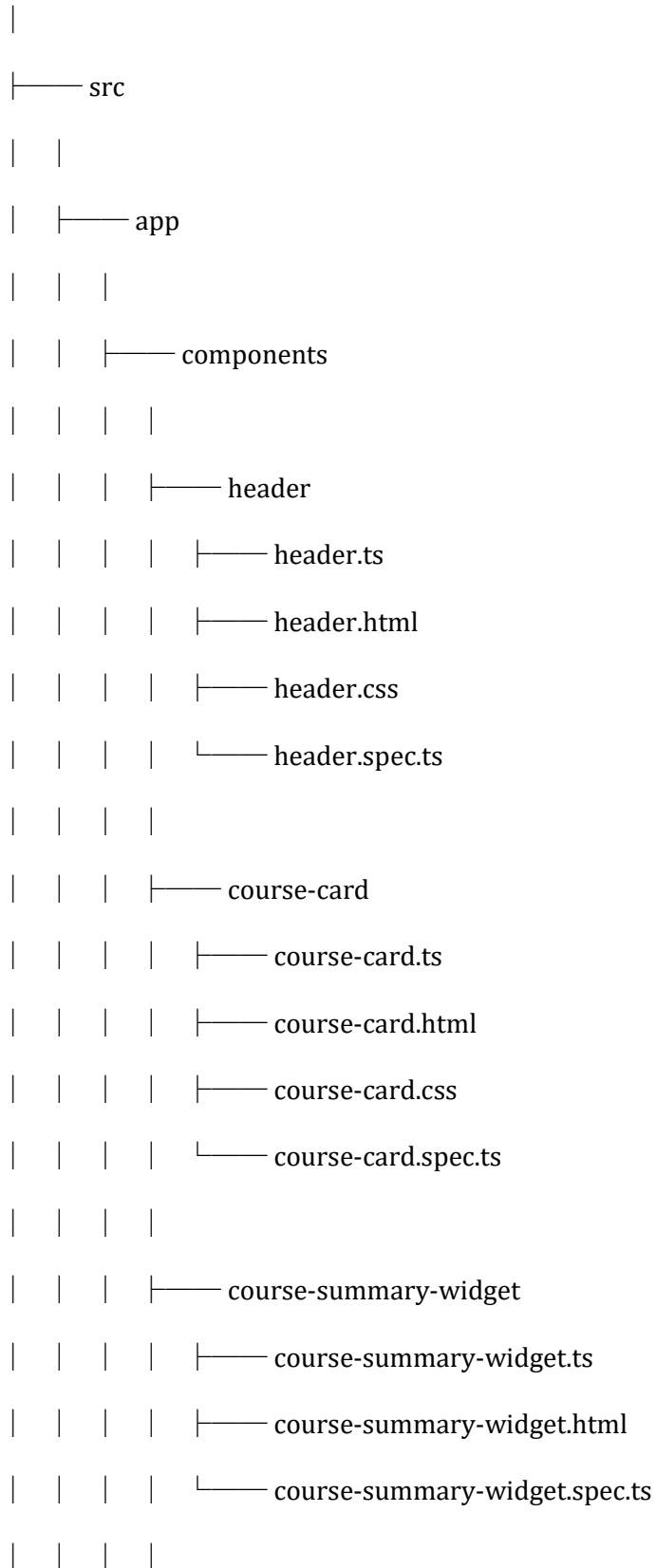
✓ student-course-portal src/app/page/course-card/course-card.spec.ts (4 tests) 138ms
✓ student-course-portal src/app/guards/auth-guard.spec.ts (1 test) 0ms
✓ student-course-portal src/app/page/course-list/course-list.spec.ts (2 tests) 23ms

Test Files 19 passed (19)
Tests 25 passed (25)
Start at 08:12:39
Duration 1.50s (transform 613ms, setup 2.47s, import 1.22s, tests 806ms, environment 13.04s)

PASS Waiting for file changes...
press h to show help, press q to quit

```

student-course-portal




```
| | | └── notification
| | |   ├── notification.ts
| | |   ├── notification.html
| | |   └── notification.spec.ts
| | |
| | |
| | └── pages
| |   |
| |   └── home
| |     ├── home.ts
| |     ├── home.html
| |     └── home.spec.ts
| |   |
| |   └── course-list
| |     ├── course-list.ts
| |     ├── course-list.html
| |     └── course-list.spec.ts
| |   |
| |   └── course-detail
| |     ├── course-detail.ts
| |     ├── course-detail.html
| |     └── course-detail.spec.ts
| |   |
| |   └── student-profile
| |     └── student-profile.ts
```

```
| | | | └── student-profile.spec.ts
| | | |
| | | └── enrollment-form
| | | | └── enrollment-form.ts
| | | | └── enrollment-form.spec.ts
| | | |
| | | └── reactive-enrollment-form
| | | | └── reactive-enrollment-form.ts
| | | | └── reactive-enrollment-form.spec.ts
| | |
| | |
| | └── services
| | | |
| | | └── course.service.ts
| | | └── course.spec.ts
| | |
| | |
| | └── store
| | | |
| | | └── course
| | | | └── course.actions.ts
| | | | └── course.reducer.ts
| | | | └── course.selectors.ts
| | | | └── course.effects.ts
| | | |
```

- | | | └── enrollment
 - | | | └── enrollment.actions.ts
 - | | | └── enrollment.reducer.ts
 - | | | └── enrollment.selectors.ts
- | | |
- | | |
- | | └── interceptors
 - | | └── auth-interceptor.ts
 - | | └── loading-interceptor.ts
 - | | └── error-handler-interceptor.ts
- | | |
- | | |
- | | └── guards
 - | | └── auth-guard.ts
- | | |
- | | |
- | | └── directives
 - | | └── highlight.directive.ts
- | | |
- | | |
- | | └── pipes
 - | | └── credit-label.pipe.ts
- | | |
- | | |
- | | └── app.routes.ts

| | | — app.config.ts

| | | — app.ts

| | | — app.html

| |

| |

| | — assets

| |

| | — index.html

| | — main.ts

| | — styles.css

|

— db.json

— angular.json

— package.json

— tsconfig.json

— README.md